ML-Driven Black-Litterman Portfolio

Backtest Summary Report

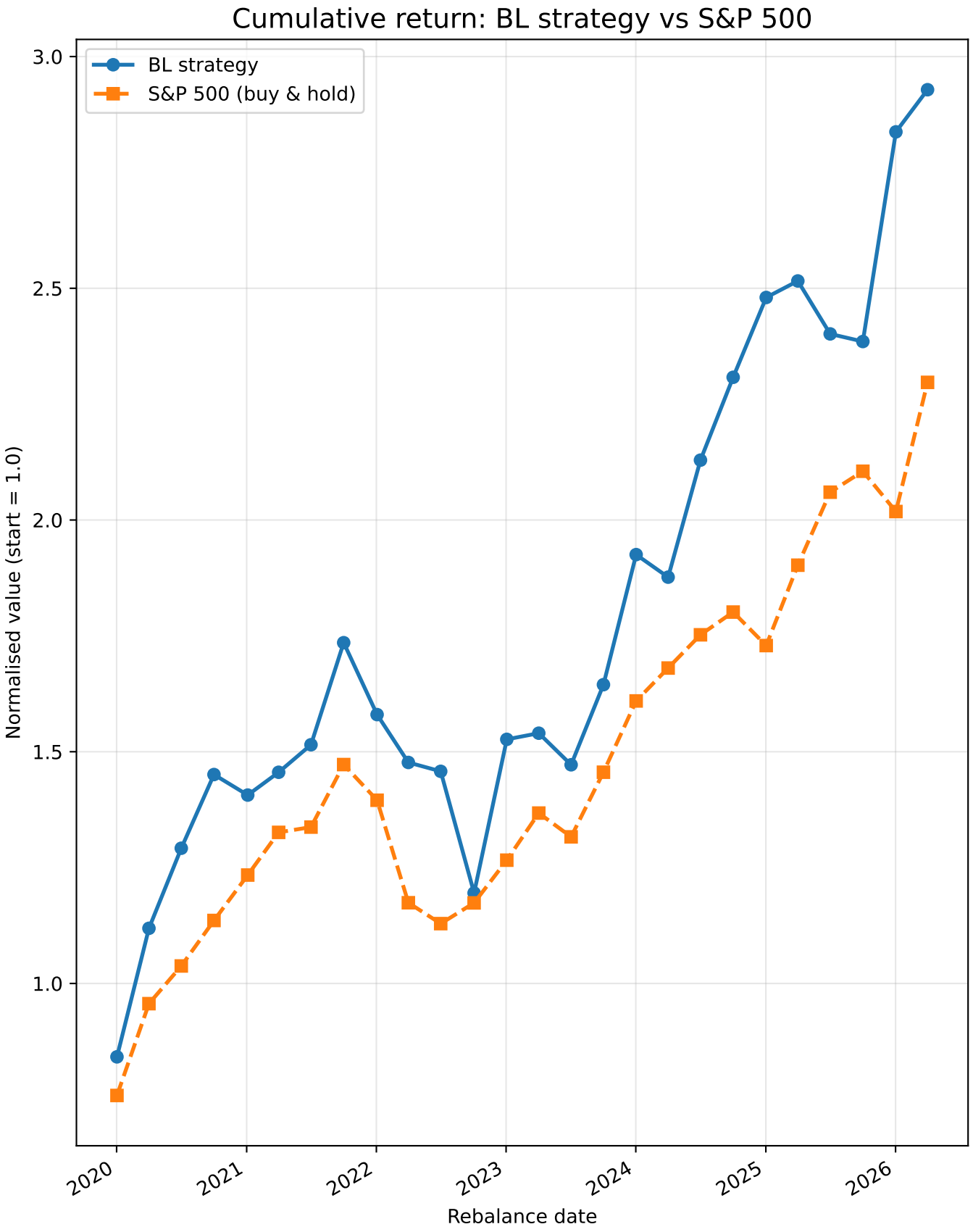
```
Performance period: 2020-01-01 → 2026-07-02   (quarterly rebalancing, 26 periods)
Risk-free rate (FRED DTB3): 3.71% annual

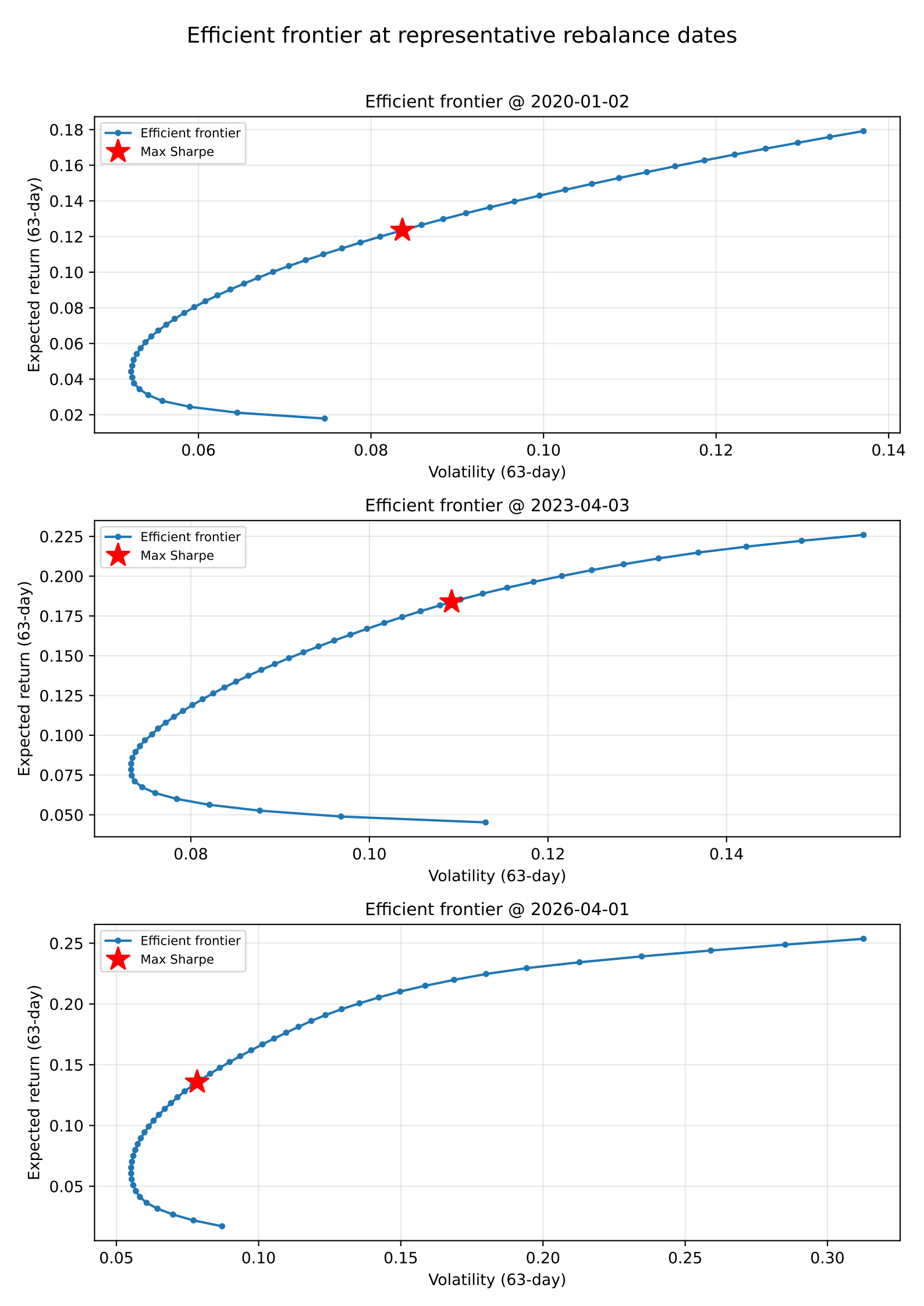
Strategy annualised return: 18.71%   vs S&P 500 (buy & hold): 13.65%
→ OUTperforming by 5.06% annualised.
Strategy Sharpe: 0.676   vs benchmark: 0.570
```

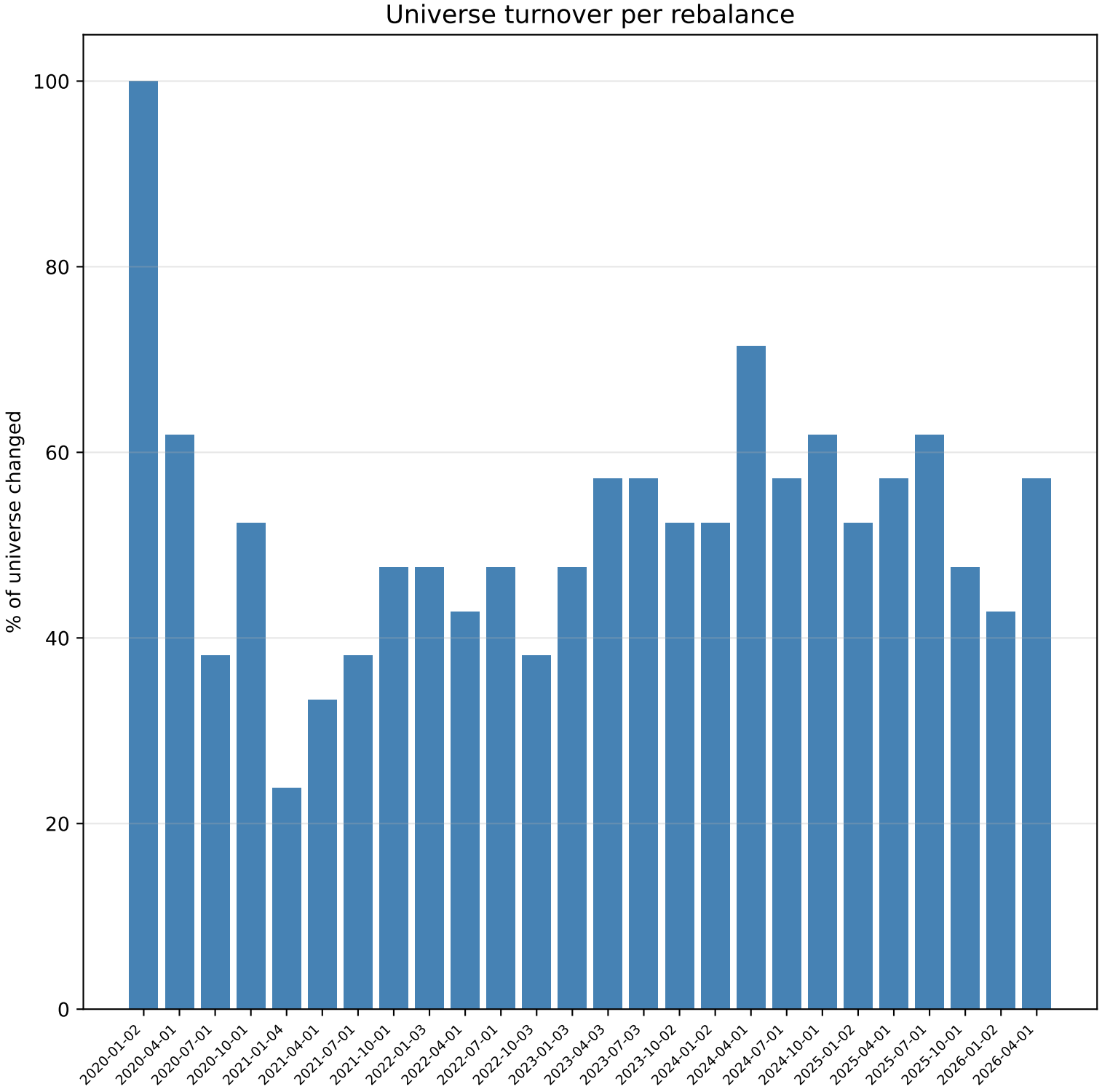
	Strategy	S&P 500 (B&H)
eturn	18.71%	13.65%
olatility	24.45%	19.66%
	0.676	0.570
n	-31.12%	-23.31%
	0.601	0.585
	-14.01%	-13.19%
	-16.81%	-20.01%

Performance & risk metrics (periods_per_year = 4)

Notes / limitations (see 08_EVALUATION_METRICS.md §7):
- Survivorship bias: current S&P 500 list applied retroactively.
- Rolling 10-year window forgets old crises (2008 GFC, 2020 COVID) as they age out.
- Market cap reconstruction approximates shares as constant.
- SVR tuned once on a sample; sector classification held fixed; overlapping 63-day targets shrink the effective sample size.
- Research scaffold — illustrative, not a profitable strategy.


Cumulative return: BL strategy vs S&P 500
BL strategy
S&P 500 (buy & hold)
Normalised value (start = 1.0)
3.0
2.5
2.0
1.5
1.0
2020
2021
2022
2023
2024
2025
2026
Rebalance date


Efficient frontier at representative rebalance dates
Efficient frontier @ 2020-01-02
Efficient frontier
Max Sharpe
Expected return (63-day)
Volatility (63-day)
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.06
0.08
0.10
0.12
0.14
Efficient frontier @ 2023-04-03
Efficient frontier
Max Sharpe
Expected return (63-day)
Volatility (63-day)
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.08
0.10
0.12
0.14
Efficient frontier @ 2026-04-01
Efficient frontier
Max Sharpe
Expected return (63-day)
Volatility (63-day)
0.25
0.20
0.15
0.10
0.05
0.05
0.10
0.15
0.20
0.25
0.30




```
ML quality diagnostics (full universe, before selection):
   Mean Information Coefficient : -0.0051   (>0.05 weakly predictive, >0.1 good)
   Mean directional accuracy    : 54.98%   (50% = random)

Total transaction cost over backtest : 4.02% of portfolio value (cumulative drag).
```